

HOME ASSISTANT

RING PERSON & FACE RECOGNITION

Requirements, Architecture, Prerequisites, Local Mac Test Plan & Dependencies

Project: Residential Home Automation / Security Intelligence

Document Status	Baseline Requirements - Phase 1
Primary Camera	Ring Front Door
Home Automation Platform	Home Assistant OS on Raspberry Pi 3
Primary AI / NVR Platform	Frigate on a separate compute host
Initial Outcome	Known-person / unknown-person identification and Home Assistant notification

1. Document Purpose

This document defines the requirements for adding local person detection and face recognition to the existing Home Assistant security setup. The system must use the Ring Front Door camera as the first camera, identify a detected person when possible, map that identity to a household relationship category, and make the result available to Home Assistant for notifications and future automations.

2. Goal

Primary goal: When the Front Door Ring camera sees a person, the system should determine whether the face matches an enrolled known person. If a confident match exists, Home Assistant should receive the person's identity and relationship category; otherwise the event must be treated as Unknown.

- Example: "Chandra detected at Front Door - Family."
- Example: "John detected at Front Door - Neighbor."
- Example: "Unknown person detected at Front Door."
- The system must never infer family/friend/neighbor from appearance. Relationship is a user-maintained mapping from a recognized name to a category.

3. Current State

Component	Current State	Notes
Home Assistant	Running	Home Assistant OS on Raspberry Pi 3
Ring integration	Configured	Front, Front Door, Garden, Lawn, Living Room, Road Side, Side
Front Door motion	Verified	event.front_door_motion
Front Door ding	Verified	event.front_door_ding
Mobile notification	Verified	notify.mobile_app_pk_world
Doorbell automation	Created	Front Door Doorbell Notification
Motion automation	Created/Test phase	Front Door Motion Notification with 5-minute cooldown
MQTT	Installed	Mosquitto Broker already present
Frigate	Not installed	Should NOT be hosted on Raspberry Pi 3
Face recognition	Not configured	Requires separate compute and enrolled face library
Ring video bridge	Not configured for AI	Ring-MQTT/go2rtc path is proposed for RTSP access

4. Scope

4.1 Phase 1 - In Scope

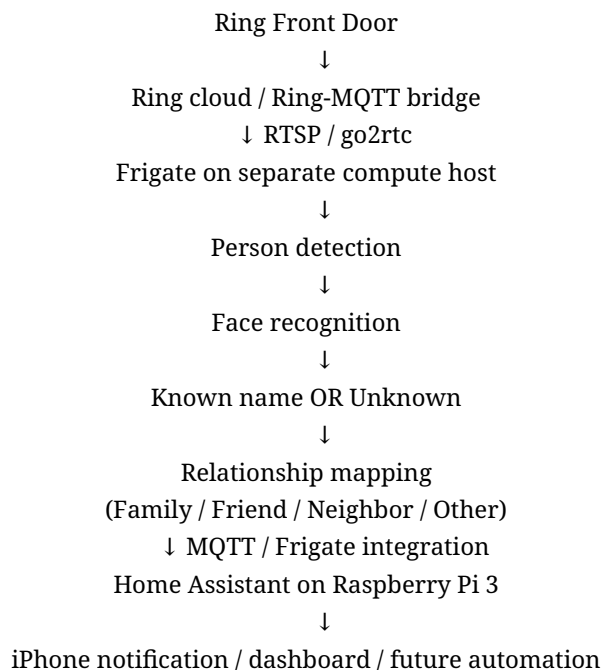
- Use only the Ring Front Door camera initially.
- Obtain a video stream suitable for local AI processing.
- Detect a person before attempting face recognition.
- Recognize enrolled known people when confidence is high enough.
- Return Unknown when the match is uncertain or no enrolled face matches.

- Maintain a manual relationship mapping: Family, Friend, Neighbor, Other Known.
- Publish results to Home Assistant for notifications and dashboard use.
- Keep Home Assistant orchestration on the Raspberry Pi 3; run video AI elsewhere.
- Test the pipeline locally on a Mac before choosing/commissioning permanent AI hardware.

4.2 Initially Out of Scope

- Running Frigate or face recognition on the Raspberry Pi 3.
- Continuous AI processing of all Ring cameras on day one.
- Guessing a person's relationship from facial appearance.
- Automatic door unlocking based only on facial recognition.
- Cloud-hosted face databases as the primary design.
- License-plate recognition, package recognition, or advanced behavior analytics in Phase 1.

5. Target Architecture



Frigate now includes native face-recognition capability. A known identity is attached to a detected person as a sub-label. This reduces the need for a separate face-recognition product for the baseline design.

6. Functional Requirements

ID	Requirement	Definition
FR-01	Video ingestion	The system shall make the Ring Front Door video available to the AI host in a format Frigate can consume.
FR-02	Person detection	The system shall detect a person before attempting face recognition.
FR-03	Face recognition	The system shall compare a usable detected face with enrolled known faces.

FR-04	Unknown handling	If recognition confidence is below the configured threshold, the result shall be Unknown.
FR-05	Identity grouping	Each known identity shall have a manually maintained relationship category.
FR-06	Home Assistant publication	Recognized identity / Unknown status shall be available to Home Assistant.
FR-07	Notification	Home Assistant shall be able to notify the configured iPhone when a person is detected.
FR-08	Event evidence	A snapshot or event reference should be retained long enough to validate recognition decisions.
FR-09	Training workflow	The user shall be able to add and refine known faces using a local face library.
FR-10	Fallback	If face recognition is unavailable, person detection shall still produce an Unknown-person event rather than suppressing the security event.

7. Non-Functional Requirements

- Local-first processing: facial embeddings and identity data should remain on local infrastructure.
- Fail-safe behavior: uncertain identities must become Unknown, not an incorrect confident identity.
- Home Assistant availability must not depend on the AI host; normal HA automations should continue if Frigate is down.
- The AI workload must not materially degrade the Raspberry Pi 3.
- The design must allow later expansion from Front Door to other Ring cameras.
- Credentials and Ring authentication tokens must not be committed to Git or stored in plain-text project files.
- Notifications should include a confidence-aware identity only when the configured recognition threshold has been met.

8. Prerequisites Before Implementation

Area	Prerequisite	Status/Action
Home Assistant	Running and reachable; Ring integration and iPhone notifications already validated.	Required before end-to-end production test
MQTT Broker	Mosquitto available. Frigate and Home Assistant must use the same MQTT broker for the HA integration.	Required before end-to-end production test
Ring account	Valid Ring account; MFA/authentication may be required by Ring-MQTT.	Required before end-to-end production test
Ring video access	A working Ring-MQTT/go2rtc RTSP stream or equivalent supported stream must be proven before AI tuning.	Required before end-to-end production test

Separate compute	A machine other than the Pi 3 is required for production Frigate/AI.	Required before end-to-end production test
Network	AI host and Home Assistant must have reliable LAN connectivity; static/DHCP-reserved addresses recommended.	Required before end-to-end production test
Training data	Initial clear, front-facing face photos for each person to be enrolled.	Required before end-to-end production test
Privacy decision	Define exactly which people are permitted to be enrolled and how long event images are retained.	Required before end-to-end production test

9. Local Testing on a Mac

Local Mac testing should be done in two stages. The first validates the architecture without depending on a live Ring stream. The second validates the full Ring-to-Frigate-to-Home-Assistant path.

9.1 Mac Hardware Check

- Identify whether the Mac is Intel or Apple Silicon (M1/M2/M3/M4).
- Allocate at least 4-8 GB of Docker memory for a basic one-camera Frigate test.
- Keep sufficient free disk space for containers, temporary recordings, snapshots and model files.
- Use wired Ethernet where possible for repeatable video-stream testing.

Important Apple Silicon limitation: Current Frigate documentation states that native face recognition requires AVX + AVX2 CPU instructions. Apple Silicon does not provide x86 AVX/AVX2. Frigate does support Apple Silicon object detection through a host-side ZMQ detector client, but that does not remove the documented AVX/AVX2 requirement for native face-recognition enrichment. Therefore an Apple Silicon Mac is suitable for validating streaming, MQTT, Home Assistant integration and person detection, but native Frigate face recognition should be treated as unsupported for the local Mac test unless Frigate's requirements change.

9.2 Stage A - Local Core Test (No Live Ring Required)

1. Install Docker Desktop.
2. Run an MQTT broker locally or connect to the existing Home Assistant Mosquitto broker.
3. Run Frigate in Docker.
4. Use a test RTSP source or looped sample video containing people to validate camera ingestion.
5. Confirm Frigate detects person objects.
6. Confirm MQTT events are produced and visible to Home Assistant.
7. Confirm Home Assistant can trigger a test notification from a Frigate person event.
8. If the Mac is an Intel model with AVX/AVX2, enable Frigate face recognition and create a small Face Library.

9.3 Stage B - Ring End-to-End Test

9. Run/configure Ring-MQTT in a Linux Docker environment and authenticate the Ring account.
10. Expose/verify the Front Door RTSP stream through Ring-MQTT/go2rtc.
11. Open the RTSP stream in VLC or ffplay before connecting it to Frigate.
12. Add the RTSP stream to Frigate as the Front Door camera.

13. Confirm person detection from a real Ring event.
14. Confirm Frigate -> MQTT -> Home Assistant event flow.
15. On supported face-recognition hardware, enroll one known person and test Known vs Unknown behavior.
16. Only after accuracy is acceptable, enable user-facing identity notifications.

10. Software Dependencies

Dependency	Purpose	Version Guidance	Required?
Docker Desktop	Required for local Mac containers	Latest stable supported by the Mac	Local development host
Frigate	Core NVR/object detection/face recognition	Current stable	Production AI service; Docker-based
Mosquitto / MQTT	Event bus between Frigate and Home Assistant	Existing HA Mosquitto is acceptable	Required by Frigate HA integration
Home Assistant Frigate Integration	Expose Frigate entities/events to HA	Install through HACS per Frigate docs	Required for HA integration
Ring-MQTT	Bridge Ring devices/video to MQTT/RTSP	Current supported Docker/add-on release	Proposed Ring video bridge
go2rtc	Low-latency stream/gateway component	Bundled/used by Ring-MQTT and Frigate workflows	Video transport
FFmpeg	Decode/transform camera streams	Provided inside Frigate container	Video ingestion/processing
VLC or ffmpeg	Optional test client	Any current version	Verify RTSP before Frigate
Apple Silicon detector client	Optional; Apple Silicon Macs only	Per Frigate docs	Accelerates object detection via host-side ZMQ
YOLO model	Optional for Apple Silicon detector	Supported model per Frigate docs	Object detection acceleration

11. Face Library / Data Requirements

- Start with 1-5 clear front-facing images per person to establish the identity.
- Then train using good-quality faces captured by the actual Front Door camera.
- Include realistic variation over time: daylight, evening, mild pose changes and normal appearance changes.
- Avoid blurry, tiny, heavily occluded or extremely dark/overexposed faces.
- Do not over-train with many nearly identical images.
- Relationship categories are separate metadata maintained by Home Assistant or a small local mapping file/helper.

12. Suggested Relationship Mapping

recognized_name -> category

Praveen -> Family

Chandra -> Family

Neighbor_A -> Neighbor

Friend_A -> Friend
Contractor -> Other Known
Unknown -> Unknown

13. Home Assistant Automation Behavior

Person detected

```
|
|-- known face above threshold
|   |
|   |-- lookup relationship
|   |-- notify: "<Name> (<Category>) detected at Front Door"
|
|-- no reliable face match
|
|-- notify: "Unknown person detected at Front Door"
```

Initial identity automation should be notification-only. Door locks, garage doors, alarms or other security-critical actuators should not be controlled solely by face recognition.

14. Test Cases & Acceptance Criteria

Test	Scenario	Pass Criteria
T01	RTSP reachable	Front Door stream opens reliably in a test player.
T02	Person detection	A person walking to the door produces a person event.
T03	Non-person filtering	Routine motion without a person does not produce an identity result.
T04	Known person	An enrolled person is recognized under normal daylight conditions.
T05	Unknown person	An unenrolled person is reported as Unknown.
T06	Low-confidence handling	A weak/partial face is not forced into a known identity.
T07	HA event flow	Frigate event is visible in Home Assistant through MQTT/integration.
T08	Notification	iPhone receives the intended known/unknown notification.
T09	AI host offline	Existing Home Assistant and Ring ding/motion automations continue functioning.
T10	Privacy	Face database and credentials are stored only on approved local systems.

15. Production Hardware Direction

The production AI host should be selected after the Mac proof-of-concept. A modern Intel mini-PC is a strong fit because Frigate supports Intel OpenVINO for object detection and can use Intel graphics for AI

enrichments. This also avoids placing video/AI load on the Raspberry Pi 3. Exact sizing should be based on the number of cameras ultimately enabled, desired retention, stream resolution and whether the larger face-recognition model is required.

16. Implementation Phases

Phase	Outcome
Phase 0 - Baseline	Keep current Ring ding/motion Home Assistant automations working.
Phase 1 - Mac POC	Validate Docker, MQTT, Frigate, stream ingestion and person detection.
Phase 2 - Ring stream	Add Ring-MQTT/go2rtc and prove Front Door live video ingestion.
Phase 3 - Identity	Enable face recognition on supported hardware; build the face library.
Phase 4 - HA intelligence	Add identity -> relationship mapping and notifications.
Phase 5 - Production host	Move Frigate/AI to dedicated hardware.
Phase 6 - Expansion	Add other Ring cameras selectively after performance and accuracy are proven.

17. Immediate Next Steps

17. Confirm the Mac model/CPU: Intel vs Apple Silicon.
18. Install/confirm Docker Desktop.
19. Create a dedicated local project folder for Frigate configuration and test media.
20. Decide whether the Mac will connect to the existing Home Assistant Mosquitto broker or use a temporary local broker.
21. Bring up Frigate with one test camera/video source.
22. Verify person detection and MQTT flow before introducing Ring-MQTT.
23. Then connect the real Front Door Ring stream.
24. Choose permanent AI hardware only after the proof-of-concept is stable.

18. Research Basis / Current Platform Notes

- **Frigate Face Recognition:** https://docs.frigate.video/configuration/face_recognition/
- **Frigate Installation:** <https://docs.frigate.video/frigate/installation/>
- **Frigate Home Assistant Integration:** <https://docs.frigate.video/integrations/home-assistant/>
- **Frigate Apple Silicon Object Detector:** https://docs.frigate.video/configuration/object_detectors/
- **Frigate Recommended Hardware:** <https://docs.frigate.video/frigate/hardware/>
- **Home Assistant Ring Integration:** <https://www.home-assistant.io/integrations/ring>
- **Ring-MQTT Home Assistant Add-on:** <https://github.com/tsightler/ring-mqtt-ha-addon>